Nested Clustered Optimization Is One End of a Schur Bridge, and the Interior Is Sometimes Provably Better

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Abstract

Nested clustered optimization allocates within each cluster from the cluster's own covariance block and then across the resulting cluster portfolios. Block inversion says the unconstrained minimum-variance portfolio has the same two-tier shape, with each block replaced by its Schur complement against every other asset. Conditioning instead on one knot from each other cluster truncates the conditioning set in the manner of a Vecchia approximation, and we give the rank-one model of cross-cluster dependence under which it is exact. Damping the complement by $\gamma \in [0, 1]$ then gives a bridge with nested clustered optimization at $\gamma = 0$ and the global optimum at $\gamma = 1$, with no linear solve larger than a cluster or the number of clusters. Under estimation error the optimal γ can be strictly interior and full coupling can remain optimal, and we give the local theorem at the minimum-variance end with exact examples of both, including a symmetric family in which the optimum is a closed form.

1 Introduction

Two-stage cluster methods partition the assets, allocate inside each cluster, and then allocate across clusters. In nested clustered optimization, NCO [1], the inner step optimizes each cluster on its own covariance block and the outer step optimizes across the resulting cluster portfolios. The outer problem has one dimension per cluster, which is the source of the method's numerical stability.

The inner step reads only the cluster's own block. The only channel by which one cluster's weights respond to another cluster is its scalar budget.

Schur-complementary allocation [2] removes the same blind spot from hierarchical risk parity [3]. At each split of a bisection tree it replaces a block by its Schur complement against the sibling block, damped by $\gamma \in [0, 1]$, and carries a companion vector alongside. The result is a bridge with hierarchical risk parity at $\gamma = 0$ and minimum variance at $\gamma = 1$.

This note builds the corresponding bridge for a flat partition. The left end is NCO and the right end is the unconstrained global minimum-variance portfolio. The obstacle is cost, since the complement of a cluster against everything else needs an inverse of nearly full size.

Spatial statistics meets the same obstacle by conditioning on a few chosen indices rather than on everything [4]. The conditionals there are Schur complements, so the device transfers directly.

We borrow its vocabulary. One member of each cluster is its *knot*, as in the low-rank spatial models of Banerjee et al. [5] that Katzfuss and Guinness [6] place inside the Vecchia framework. The other clusters' knots are a cluster's *conditioning set*. We state the model under which conditioning on that set loses nothing.

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Section 7 asks where to sit on the bridge when the covariance is estimated. Part of the analysis of the tree in [7] transfers, and exact examples show that both an interior optimum and a locally optimal full coupling occur. Section 8 treats singular covariances, where the parameterization by γ itself becomes singular.

All exactness statements concern the unconstrained, fully invested, long-short problem. With box or long-only constraints no block decomposition of the optimum is exact, here or on the tree.

2 Setup

Let $\Sigma \succ 0$ be the covariance of n assets partitioned into clusters $I_1, \dots, I_k$. Cluster i has a knot $p_i \in I_i$ and remaining members $J_i = I_i \setminus \{p_i\}$. Write $P = \{p_1, \dots, p_k\}$ and $P_{-i} = P \setminus \{p_i\}$. For index sets X, Y write Σ_{XY} for the corresponding block, Σ_{ii} for the block of cluster i, and $-i$ for the complement of I_i.

A two-stage method chooses a portfolio v_i for each cluster, supported on I_i with $\mathbf{1}^\top v_i = 1$, and then budgets across the clusters. With V the $n \times k$ matrix of cluster portfolios,

$$a \;=\; \mathrm{alloc}(V^\top \Sigma V), \qquad w \;=\; Va. \tag{1}$$

NCO takes v_i to be the minimum-variance portfolio of Σ_{ii} and minimum variance for the outer allocation.

3 Block inversion

Fix a vector u and a cluster i. Block inversion of Σ on the partition $(I_i, -i)$ gives [2, App. A]

$$\left(\Sigma^{-1}u\right)_{I_i} \;=\; \left(\Sigma^{c}_{ii}\right)^{-1} b_i, \tag{2}$$

where

$$\Sigma^{c}_{ii} = \Sigma_{ii} - \Sigma_{i,-i}\Sigma^{-1}_{-i,-i}\Sigma_{-i,i}, \qquad b_i = u_{I_i} - \Sigma_{i,-i}\Sigma^{-1}_{-i,-i}u_{-i}. \tag{3}$$

For a positive-definite Q and a vector b we call $Q^{-1}b$ a *generalized minimum-variance direction*. Up to scale it minimizes $x^\top Q x$ subject to $b^\top x = 1$, which is ordinary minimum variance only when $b = \mathbf{1}$.

With $u = \mathbf{1}$, the blocks of the global minimum-variance direction $\Sigma^{-1}\mathbf{1}$ are therefore generalized minimum-variance directions, one per cluster, on the cluster's covariance conditional on everything else. There is no cross-cluster optimizer in the global optimum. The cluster totals are whatever the stacked directions sum to, and every piece of cross-cluster information enters through the conditioning.

NCO has the same shape with the conditioning removed. Restoring it costs an inverse of size $n - |I_i|$ per cluster, which is what a two-stage method exists to avoid.

4 The gateway model

Definition 1 (Gateway model)**.** *For every cluster j, the residual of the remaining members J_j after linear regression on their own knot is uncorrelated with every asset outside I_j. Equivalently the cross-cluster residual*

$$R_j(p_j) \;=\; \Sigma_{J_j,-j} - \Sigma_{J_j p_j}\, \sigma_{p_j}^{-2}\, \Sigma_{p_j,-j} \tag{4}$$

is zero.

A member may be correlated with anything, but its correlation with other clusters passes entirely through its knot. The knot is the cluster's only gateway. Each cross-cluster block of Σ then has rank at most one.

This is a strong assumption, and clustering alone does not imply it. It asks that member residuals be uncorrelated both with the other knots and with the other clusters' member residuals.

Proposition 1 (Knots are sufficient)**.** *Under the gateway model, for each cluster i and every u,*

$$\Sigma^{c}_{ii} \;=\; \Sigma_{ii} - \Sigma_{i,P_{-i}}\,\Sigma^{-1}_{P_{-i}P_{-i}}\,\Sigma_{P_{-i},i}, \qquad b_i \;=\; u_{I_i} - \Sigma_{i,P_{-i}}\,\Sigma^{-1}_{P_{-i}P_{-i}}\,u_{P_{-i}}. \tag{5}$$

Proof. The matrix $\Sigma_{i,-i}\Sigma^{-1}_{-i,-i}$ holds the coefficients of the linear regression of cluster i on all other assets. Split the regressors into the other knots P_{-i} and the other remaining members J_{-i}, and write $J_{-i} = \beta\, P_{-i} + \varepsilon$ for the block-diagonal regression of each set of remaining members on its own knot. The map from (P_{-i}, J_{-i}) to (P_{-i}, ε) is invertible, so the regression may be run on the latter pair. Under the gateway model ε is uncorrelated with cluster i and with P_{-i}, so its coefficients vanish and the coefficients on P_{-i} are $\Sigma_{i,P_{-i}}\Sigma^{-1}_{P_{-i}P_{-i}}$. Substituting into (3) gives (5), with u_{-i} contributing only through its P_{-i} entries. □

Conditioning cluster i on the other knots is the same as conditioning it on everything. The inverse in (5) is $(k-1)\times(k-1)$.

Remark 1 (Relation to Vecchia approximations)**.** *A Vecchia approximation orders the variables and replaces each conditioning set by a small subset of the predecessors, which defines one acyclic approximate joint law [4, 6]. The conditionals are Schur complements. Replacing $-i$ by P_{-i} in* (3) *is the same truncation of a conditioning set, but it is applied to the full conditional of every cluster at once. Those conditionals are mutually cyclic and no approximate joint law is formed, so the recipe is Vecchia-inspired rather than a Vecchia approximation.*

The gateway model itself does have an exact Vecchia form.

Corollary 1 (Gaussian factorization)**.** *If the returns r are jointly Gaussian, the gateway model holds if and only if*

$$f(r) \;=\; f(r_P)\,\prod_{i=1}^{k} f\big(r_{J_i} \mid r_{p_i}\big). \tag{6}$$

Proof. Under the model the residuals $\varepsilon_i = r_{J_i} - \beta_i r_{p_i}$ are uncorrelated with r_P and with each other, hence independent under Gaussianity, which gives (6). Conversely (6) makes ε_i independent of every asset outside I_i. □

With the knots ordered first, (6) is a block-Vecchia factorization, a sparser relative of the knot-based cases collected by Katzfuss and Guinness [6]. The covariance results in this note do not need Gaussianity.

Remark 2 (Testing the model)**.** *Regressing each remaining member on its own knot and the other knots, and checking that the other-knot coefficients vanish, does not test the model. Member residuals can be orthogonal to every knot and still be correlated with each other across clusters, and the sufficiency of the knots then fails. The model holds exactly when $R_j(p_j) = 0$ for every j, which can be checked from the sample covariance before any portfolio is built.*

Remark 3 (Measuring the loss and choosing knots)**.** *The raw size of R_j does not measure what is lost when the model fails. A residual with variance δ^2 and covariance $c\delta$ with an outside asset has $R = c\delta \to 0$, while the Schur correction it carries, c^2, does not move. The loss for cluster i is the omitted correction*

$$\Delta_i \;=\; \Sigma_{i,-i}\Sigma_{-i,-i}^{-1}\Sigma_{-i,i} - \Sigma_{i,P_{-i}}\Sigma_{P_{-i}P_{-i}}^{-1}\Sigma_{P_{-i},i} \;\succeq\; 0, \tag{7}$$

the part of cluster i explained by the other assets beyond their knots. A scale-free screen for a candidate knot $p \in I_j$ is the largest canonical correlation between the residual of $I_j \setminus \{p\}$ on p and the assets outside I_j. Both need a large inverse, but they are diagnostics computed once and are not part of the allocation.

Remark 4 (A symmetric factor model)**.** *Suppose instead that every member of cluster j loads on a latent cluster factor, $r_m = \lambda_m f_j + \eta_m$, with idiosyncratic terms uncorrelated with each other and with the factors. Then $R_j(p) = (1-\rho_p)\,\lambda_{J_j}\,\mathrm{Cov}(f_j, r_{-j})$, where $\rho_p = \lambda_p^2\,\mathrm{Var}(f_j)/\sigma_p^2$ is the share of the knot's variance explained by the factor. The gateway model holds to the extent the knot is the factor. A knot with a high ρ_p is then the natural choice, and the member most aligned with the cluster's first principal component approximates it.*

5 The bridge

Following [2], damp the conditioning. For $\gamma \in [0,1]$ define the *conditioned pair* of cluster i,

$$\begin{aligned} Q_i(\gamma) &= \Sigma_{ii} - \gamma\,\Sigma_{i,P_{-i}}\Sigma_{P_{-i}P_{-i}}^{-1}\Sigma_{P_{-i},i}, \\ b_i(\gamma) &= u_{I_i} - \gamma\,\Sigma_{i,P_{-i}}\Sigma_{P_{-i}P_{-i}}^{-1}u_{P_{-i}}, \end{aligned} \tag{8}$$

and the direction $d_i(\gamma) = Q_i(\gamma)^{-1}b_i(\gamma)$. Let D_γ be the $n \times k$ matrix with $d_i(\gamma)$ in column i, supported on I_i. The bridge portfolio is

$$w_\gamma \;\propto\; D_\gamma\big(D_\gamma^\top \Sigma D_\gamma\big)^{-1}D_\gamma^\top u. \tag{9}$$

It solves, on the span of the k directions, the problem that $\Sigma^{-1}u$ solves on the whole space. Since $b_i(\gamma) = u_{I_i} - \gamma a_i$ with $a_i = \Sigma_{i,P_{-i}}\Sigma_{P_{-i}P_{-i}}^{-1}u_{P_{-i}}$ fixed, either b_i vanishes identically, which happens exactly when $u_{I_i} = 0$ and $a_i = 0$, or it vanishes at no more than one γ, and there its shape has the limit $-Q_i(\gamma)^{-1}a_i \neq 0$. We use that limiting direction, which keeps the span continuous in γ; only its scale is lost, and scale does not enter (9). A cluster is dropped only when b_i vanishes identically; under the gateway model its block of $\Sigma^{-1}u$ is then zero. A cluster with $u_{I_i} = 0$ but $a_i \neq 0$ is kept, with direction $-Q_i(\gamma)^{-1}a_i$ for every γ: it carries no exposure to u and enters the outer tier only as a hedge. The columns of D_γ are then nonzero with disjoint supports, so $D_\gamma^\top \Sigma D_\gamma \succ 0$.

Two-tier reading. The right side of (9) does not change when a column of D_γ is rescaled. When every total $\mathbf{1}^\top d_i(\gamma)$ is nonzero, set $v_i = d_i/\mathbf{1}^\top d_i$. Then (9) is the two-stage method (1) with $a \propto (V^\top \Sigma V)^{-1}V^\top u$. The induced outer inputs are the $k \times k$ covariance $V^\top \Sigma V$ and the k-vector $V^\top u$, which is $\mathbf{1}$ when $u = \mathbf{1}$. The unnormalized form (9) has no singularity when a cluster total crosses zero.

Proposition 2 (The two ends of the bridge)**.**

1. *At $\gamma = 0$ with $u = \mathbf{1}$ the recipe is NCO with minimum variance at both tiers.*
2. *Under the gateway model, for every cluster with $b_i(1) \neq 0$ the column $d_i(1)$ of D_1 is the block $(\Sigma^{-1}u)_{I_i}$, every cluster with $b_i(1) = 0$ has a zero block, and $w_1 \propto \Sigma^{-1}u$.*

Proof. At $\gamma = 0$ the pair is $(\Sigma_{ii}, \mathbf{1})$, so $d_i = \Sigma_{ii}^{-1}\mathbf{1}$ has a positive total, v_i is the minimum-variance portfolio of the block, and the outer tier is minimum variance across the cluster portfolios. For the second claim, (2) and proposition 1 give $(\Sigma^{-1}u)_{I_i} = Q_i(1)^{-1}b_i(1)$, which is $d_i(1)$ when $b_i(1) \neq 0$ and zero otherwise. So $\Sigma^{-1}u$ is the sum of the columns with $b_i(1) \neq 0$, which lies in the column span of D_1 whatever limiting direction occupies a column with $b_i(1) = 0$. Then $D_1(D_1^\top \Sigma D_1)^{-1}D_1^\top u = D_1(D_1^\top \Sigma D_1)^{-1}D_1^\top \Sigma D_1 \alpha = D_1\alpha$ for the α with $\Sigma^{-1}u = D_1\alpha$. □

γ	Method
0	NCO [1]
$(0, 1)$	the interior of the bridge
1	unconstrained global minimum variance, under the gateway model

Table 1: The bridge with $u = \mathbf{1}$ and minimum variance at both tiers.

Scope. The right end is the unconstrained, fully invested, long-short solution. No condition on cluster totals is needed for (9). A global solution can hold a self-financing position inside a cluster with zero net weight. The two-tier reading fails there, because no cluster budget times a normalized cluster portfolio represents it, and (9) does not.

Remark 5 (Other objectives)**.** *Proposition 2 is stated for a direction $\Sigma^{-1}u$. With $u = \mu$ the direction $\Sigma^{-1}\mu$ maximizes the Sharpe ratio. Its fully invested multiple does so only when $\mathbf{1}^\top\Sigma^{-1}\mu > 0$. When that total is negative the fully invested multiple has the opposite orientation and the lowest signed Sharpe ratio.*

Mean–variance problems with a budget constraint have solutions that combine $\Sigma^{-1}\mu$ and $\Sigma^{-1}\mathbf{1}$, and each part is covered separately. Risk parity is not of this form. The recipe still runs with a risk-parity outer tier, but nothing is exact at $\gamma = 1$.

Remark 6 (Rules that read only a covariance)**.** *An inner rule that cannot take a constraint vector can be applied after a change of variables. With $y = b_i \circ x$ the constraint $b_i^\top x = 1$ becomes $\mathbf{1}^\top y = 1$ and the covariance becomes $M = \operatorname{diag}(b_i)^{-1}Q_i \operatorname{diag}(b_i)^{-1}$. Apply the rule to M and map back by $x = y/b_i$, coordinate by coordinate. For minimum variance this returns $Q_i^{-1}b_i$. It requires b_i to have no zero entry. Applying minimum variance to M without mapping back returns $\operatorname{diag}(b_i)Q_i^{-1}b_i$, which is a different portfolio.*

Remark 7 (Relation to the tree)**.** *Schur complements compose: the complement of a complement is the complement against the union. So at $\gamma = 1$ the recursion of [2], run down any tree whose leaves are the clusters and stopped there, hands each cluster the pair* (3)*. The flat construction and the iterated one agree at the right end. They differ in the interior, because the tree damps at every split and the damping compounds along the path, while* (8) *damps each cluster once against all the other knots.*

6 Structure under the gateway model

The gateway model gives both the outer covariance and the cluster directions a simple form.

Proposition 3 (Knot covariance plus a ridge)**.** *Under the gateway model, let v_i be any portfolios supported on their clusters. Write $\beta_i = \Sigma_{J_i p_i}/\sigma^2_{p_i}$ for the member betas, $E_i = \Sigma_{J_i J_i} - \beta_i \beta_i^\top \sigma^2_{p_i}$ for the residual covariance, and $\kappa_i = v_{i,p_i} + \beta_i^\top v_{i,J_i}$ for the cluster portfolio's beta to its knot. Then*

$$V^\top \Sigma V \;=\; K\,\Sigma_{PP}\,K + D, \qquad K = \operatorname{diag}(\kappa_i), \quad D = \operatorname{diag}\big(v_{i,J_i}^\top E_i\, v_{i,J_i}\big). \tag{10}$$

Proof. Under the gateway model $\Sigma_{J_i,l} = \beta_i \Sigma_{p_i,l}$ for every asset l outside I_i. Applying this on both sides gives $v_i^\top \Sigma_{I_i I_l} v_l = \kappa_i \kappa_l \Sigma_{p_i p_l}$ for $i \neq l$. On the diagonal, the knot is uncorrelated with the member residual, so $v_i^\top \Sigma_{ii} v_i = \kappa_i^2 \sigma^2_{p_i} + v_{i,J_i}^\top E_i v_{i,J_i}$. □

The outer problem sees the knot covariance, scaled by the cluster portfolios' knot betas, plus their residual variances on the diagonal. The diagonal term raises the smallest eigenvalue of the outer covariance by at least the smallest residual variance. The remaining members supply a ridge, which is one reading of why the outer tier of NCO is stable.

Proposition 4 (Only the knots move)**.** *Under the gateway model, split the direction $d_i(\gamma)$ into its member part d_{i,J_i} and its knot exposure $\kappa_i = d_{i,p_i} + \beta_i^\top d_{i,J_i}$. Then*

$$d_{i,J_i}(\gamma) = E_i^{-1}\big(u_{J_i} - \beta_i\, u_{p_i}\big), \qquad \kappa_i(\gamma) = \frac{u_{p_i} - \gamma\, a_i}{\sigma^2_{p_i} - \gamma\, c_i}, \tag{11}$$

where $a_i = \Sigma_{p_i,P_{-i}} \Sigma^{-1}_{P_{-i}P_{-i}} u_{P_{-i}}$ and $c_i = \Sigma_{p_i,P_{-i}} \Sigma^{-1}_{P_{-i}P_{-i}} \Sigma_{P_{-i},p_i}$. The member part depends on neither γ nor any other cluster.

Proof. Under the model $\Sigma_{I_i,P_{-i}} = \binom{1}{\beta_i} \Sigma_{p_i,P_{-i}}$, so $Q_i(\gamma) = \Sigma_{ii} - \gamma c_i \binom{1}{\beta_i}\binom{1}{\beta_i}^\top$ and $b_i(\gamma) = u_{I_i} - \gamma a_i \binom{1}{\beta_i}$. In the coordinates (κ, x_J) the quadratic form $x^\top Q_i x$ is $(\sigma^2_{p_i} - \gamma c_i)\kappa^2 + x_J^\top E_i x_J$ and the constraint $b_i^\top x$ is $(u_{p_i} - \gamma a_i)\kappa + (u_{J_i} - \beta_i u_{p_i})^\top x_J$. The two parts decouple, and $Q_i^{-1} b_i$ is (11). □

NCO and the global optimum hold the same member position inside every cluster, up to the cluster's scale. Moving along the bridge reprices only the knot. With $u = \mathbf{1}$ a member earns weight in proportion to $1 - \beta$, its shortfall in beta to the knot, per unit of residual variance.

Corollary 2 (A cluster is a knot and one independent asset)**.** *Take $u = \mathbf{1}$ and put $\delta_i = (\mathbf{1} - \beta_i)^\top E_i^{-1} (\mathbf{1} - \beta_i)$. Under the gateway model, replace each cluster by its knot together with one asset that is uncorrelated with everything and has variance $1/\delta_i$. Then the variance $V_0(\gamma) = w_\gamma^\top \Sigma w_\gamma$ of the normalized bridge portfolio, $\mathbf{1}^\top w_\gamma = 1$, is unchanged for every γ. So is the expected out-of-sample variance $F(\gamma, \tau)$ of section 7 under noise that perturbs only the knot covariance and preserves the gateway structure.*

Proof. By proposition 4 the direction of cluster i has knot exposure $\kappa_i(\gamma)$, total weight $\kappa_i(\gamma) + \delta_i$ and residual variance δ_i. By proposition 3 the outer tier, and the variance of the resulting portfolio, read the cluster only through these and the knot covariance. The replacement cluster has $\beta = 0$ and $E = 1/\delta_i$, which gives the same three quantities. □

A large cluster interior therefore never obscures the analysis: it enters as one scalar. If in every cluster every member has unit beta to its knot then every $\delta_i = 0$, the members carry no budget, and w_γ is the same for every γ at which the knot exposures are nonzero. With $\delta_i = 0$ in some clusters only, w_γ still moves.

For the rest of this section take $u = \mathbf{1}$ and write $S = \Sigma_{PP}$ for the knot covariance, $s_i = S_{ii}$, $h = S^{-1}\mathbf{1}$ and $\nu_i = 1/(S^{-1})_{ii}$, the variance of knot i left unexplained by the other knots. In the notation of proposition 4, $\nu_i = s_i - c_i$ and $h_i = (1 - a_i)/\nu_i$.

Proposition 5 (Effective damping)**.** *With* $r_i = \nu_i / s_i$*,*

$$\kappa_i(\gamma) = (1 - \lambda_i)\,\frac{1}{s_i} + \lambda_i\, h_i, \qquad \lambda_i = \frac{\gamma\, r_i}{1 - \gamma + \gamma\, r_i}, \qquad \textit{that is} \qquad \frac{\lambda_i}{1 - \lambda_i} = r_i\,\frac{\gamma}{1 - \gamma}. \tag{12}$$

Proof. Write the denominator of κ_i as $s_i(1-\gamma) + \gamma\nu_i$ and the numerator as $(1-\gamma) + \gamma\nu_i h_i$, and divide. □

The knot exposure interpolates between its NCO value $1/s_i$ and its minimum-variance value h_i, and λ_i is the interpolation actually achieved. A common γ does not damp the clusters equally. A knot that the other knots nearly explain has small r_i, and its exposure barely moves until γ is very close to 1.

Proposition 6 (Lost precision)**.** *Under the gateway model, suppose every* $\delta_i > 0$*. Put* $\Delta = \operatorname{diag}(\delta_i)$*,* $K_\gamma = \operatorname{diag}(\kappa_i(\gamma))$*,* $Z = \mathbf{1}^\top h + \sum_i \delta_i$ *and* $W_\gamma = \big(S^{-1} + K_\gamma \Delta^{-1} K_\gamma\big)^{-1}$*. The minimum variance is* $V_\star = 1/Z$*, and*

$$V_0(\gamma) - V_\star = \frac{L_\gamma}{Z\,(Z - L_\gamma)}, \qquad L_\gamma = (h - \kappa)^\top W_\gamma\,(h - \kappa). \tag{13}$$

Proof. By corollary 2 work with $2k$ assets of covariance $\Sigma = \operatorname{diag}(S, \Delta^{-1})$ and budget vector $(\mathbf{1}, \mathbf{1})$. The unconstrained precision is Z and its direction is $\Sigma^{-1}\mathbf{1} = (h, \delta)$. Column i of D_γ is $(\kappa_i(\gamma)e_i, \delta_i e_i)$; since $\delta_i > 0$ it is nonzero even where $\kappa_i(\gamma) = 0$, so D_γ has rank k on the whole bridge. The bridge restricts the knot exposures to $K_\gamma a$ and the member weights to Δa for an outer vector a, and its precision $1/V_0(\gamma) = (D_\gamma^\top \mathbf{1})^\top (D_\gamma^\top \Sigma D_\gamma)^{-1} D_\gamma^\top \mathbf{1}$ falls short of Z by the squared Σ-distance from (h, δ) to that subspace,

$$\min_a \big\{(h - K_\gamma a)^\top S\,(h - K_\gamma a) + (\mathbf{1} - a)^\top \Delta\,(\mathbf{1} - a)\big\},$$

because expanding $\|\Sigma^{-1}\mathbf{1} - D_\gamma a\|_\Sigma^2$ and minimizing over a gives exactly $Z - 1/V_0(\gamma)$. Substitute $a = \mathbf{1} - b$, so that $h - K_\gamma a = (h - \kappa) + K_\gamma b$. The minimum over b is attained at $b = -(K_\gamma S K_\gamma + \Delta)^{-1} K_\gamma S\,(h - \kappa)$ and equals $(h-\kappa)^\top \big[S - S K_\gamma (K_\gamma S K_\gamma + \Delta)^{-1} K_\gamma S\big](h - \kappa)$. The bracket is W_γ by the Woodbury identity, which gives L_γ, and $1/(Z - L_\gamma) - 1/Z$ is (13). □

Since $W_\gamma \succ 0$, the bridge attains the minimum variance exactly when $\kappa = h$, and

$$\kappa_i(\gamma) - h_i = \frac{(1-\gamma)\,(1 - s_i h_i)}{(1-\gamma)s_i + \gamma\nu_i}. \tag{14}$$

So if some $s_i h_i \neq 1$ then $\gamma = 1$ is the unique global minimizer of V_0 on $[0, 1]$. If every $s_i h_i = 1$ the whole bridge is already optimal. The curvature at the end is explicit as well. With $\dot\kappa_i = \kappa_i'(1) = (s_i h_i - 1)/\nu_i$ and $W_1 = \big(S^{-1} + \operatorname{diag}(h_i^2/\delta_i)\big)^{-1}$,

$$V_0''(1) = \frac{2\,\dot\kappa^\top W_1\,\dot\kappa}{Z^2}. \tag{15}$$

Equicorrelated knots. Let the k knots have unit variance and common correlation ρ, and take $u = \mathbf{1}$. Then $a = \rho(k-1)/\big(1 + (k-2)\rho\big)$ and $c = \rho\, a$, so every cluster has

$$\kappa(\gamma) = \frac{1 + (k-2)\rho - \gamma\,\rho\,(k-1)}{1 + (k-2)\rho - \gamma\,\rho^2(k-1)}.$$

For $\rho > 0$ the knot exposure falls from 1 at the NCO end to $1/\big(1 + (k-1)\rho\big)$ at the minimum-variance end, which is the scale of the equicorrelated minimum-variance weight. The more knots and the higher their correlation, the further the bridge has to travel.

Remark 8 (Degenerate partitions)**.** *With every asset its own cluster each direction is a multiple of a coordinate vector, so, with the limiting direction used wherever a $b_i(\gamma)$ vanishes, the directions span the whole space at every γ. With a single cluster there is nothing to condition on. In both cases w_γ is the minimum-variance portfolio for every γ, and no model is needed. The bridge has length only for partitions strictly between the two.*

7 Where to sit on the bridge

The cross-knot regression coefficients in (8) are estimated, and γ is how much of that estimate to trust. The analysis of the tree in [7] transfers in part. This section takes $u = \mathbf{1}$.

Let $\widehat{\Sigma}_\tau = \Sigma + \tau E$ be an estimate, where E is a random symmetric matrix whose law is unchanged by $E \mapsto -E$, and $\widehat{\Sigma}_\tau \succ 0$ on the range of τ considered. Every quantity in the recipe is computed from $\widehat{\Sigma}_\tau$. The partition and the knots are held fixed as τ varies. A data-driven choice of either, such as the knot screens of section 4, is a discrete function of $\widehat{\Sigma}_\tau$: away from its decision boundaries it is locally constant, so the differential statements below hold for noise too small to switch it, and at a switching level F can lose smoothness. The portfolio w_γ of (9) is normalized by $\mathbf{1}^\top w_\gamma = 1$, and we assume the normalizing total is bounded away from zero on the range of (γ, τ) considered; a total near zero makes w_γ large and the expansion below non-uniform in γ. Define the expected out-of-sample variance and its noiseless profile,

$$F(\gamma, \tau) = \mathbb{E}\big[w_\gamma(\widehat{\Sigma}_\tau)^\top \, \Sigma \, w_\gamma(\widehat{\Sigma}_\tau)\big], \qquad V_0(\gamma) = F(\gamma, 0). \tag{16}$$

It is even in τ. We assume F is smooth, put $G(\gamma) = \frac{1}{2}\,\partial_\tau^2 F(\gamma, 0)$, and assume the expansion $F = V_0 + \tau^2 G + O(\tau^4)$ holds in C^2 as a function of γ.

Proposition 7 (Incremental noise cost)**.** *Let $\Phi(\gamma, \tau) = F(\gamma, \tau) - F(0, \tau) - \big[V_0(\gamma) - V_0(0)\big]$. Then $\Phi(\gamma, \tau) = \gamma\, \tau^2 H(\gamma, \tau)$ for a smooth H.*

Proof. $\Phi(0, \tau) = 0$, so $\Phi = \gamma\, \Phi_1$ with Φ_1 smooth. Also $\Phi(\gamma, 0) = 0$, so $\Phi_1(\gamma, 0) = 0$, and Φ_1 is even in τ, so $\Phi_1 = \tau^2 H$. □

On the tree, under noise supported only on the top-level cross-block, the hierarchical end does not read that block, and the uncentred cost $F - V_0$ already carries the factor γ [7]. Under whole-matrix noise that fails on the tree as well, because $F(0, \tau)$ is then noisy.

NCO is different even under cross-cluster noise alone. Its outer tier reads cross-cluster covariance at $\gamma = 0$, so NCO has an estimation cost of its own, and the factor γ appears only after centring at the noisy NCO end. The centred statement holds for the general noise of this section, and the cost of moving away from NCO is still of order $\gamma\tau^2$.

Corollary 3 (Separation from NCO)**.** *If $V_0'(0) < 0$ then $\partial_\gamma F(0, \tau) < 0$ for all sufficiently small τ, and NCO is strictly suboptimal.*

Proof. By proposition 7, $\partial_\gamma F(0, \tau) = V_0'(0) + \tau^2 H(0, \tau)$. □

Proposition 8 (The minimum-variance end)**.** *Under the gateway model $V_0'(1) = 0$ and $V_0''(1) = 2\, \dot{w}^\top \Sigma\, \dot{w}$, where $\dot{w} = \partial_\gamma w_\gamma$ at $\gamma = 1$. Suppose $\dot{w} \neq 0$. Then for all sufficiently small τ the map $F(\cdot, \tau)$ has a unique stationary point near 1, a strict local minimizer, at*

$$\tilde{\gamma}(\tau) \;=\; 1 - \frac{G'(1)}{V_0''(1)}\, \tau^2 + O(\tau^4). \tag{17}$$

If $G'(1) > 0$ this local minimizer is strictly inside the bridge for small $\tau > 0$. If $G'(1) < 0$ then $\gamma = 1$ is a strict local minimizer of $F(\cdot, \tau)$ on $[0, 1]$. If $G'(1) = 0$ the second order is silent and the fourth or a higher order decides.

Suppose in addition that $\gamma = 1$ is the unique global minimizer of V_0 on $[0, 1]$ and that $F(\cdot, \tau) \to V_0$ uniformly on $[0, 1]$. Then for all sufficiently small τ these local minimizers are the global optimum over $[0, 1]$.

Proof. Since $\mathbf{1}^\top w_\gamma = 1$ we have $\mathbf{1}^\top \dot{w} = 0$ and $\mathbf{1}^\top \ddot{w} = 0$. By proposition 2, $w_1 = \Sigma^{-1}\mathbf{1}/\mathbf{1}^\top\Sigma^{-1}\mathbf{1}$, so $\Sigma w_1 \propto \mathbf{1}$. Hence $V_0'(1) = 2\dot{w}^\top\Sigma w_1 = 0$, and in $V_0''(1) = 2\dot{w}^\top\Sigma\dot{w} + 2\ddot{w}^\top\Sigma w_1$ the second term vanishes. Apply the implicit function theorem to $\partial_\gamma F(\gamma, \tau) = 0$ at $(1, 0)$, where $\partial_\gamma^2 F = V_0''(1) > 0$. This gives (17) and strict convexity of $F(\cdot, \tau)$ on a neighbourhood of 1 that does not depend on τ.

When $G'(1) < 0$ the stationary point lies beyond 1, so $F(\cdot, \tau)$ decreases up to the boundary. Under the additional hypotheses the global minimizers of $F(\cdot, \tau)$ converge to 1, so for small τ they lie in that neighbourhood. □

When every $\delta_i > 0$ the uniqueness hypothesis is supplied by proposition 6, unless the whole bridge is already optimal. It remains a hypothesis when some $\delta_i = 0$, since gateway exactness alone does not exclude another $\gamma < 1$ that produces the same portfolio.

Remark 9 (When the gateway model fails)**.** *If some $\Delta_i \neq 0$ in (7), then w_1 is not the minimum-variance portfolio and $V_0'(1)$ is in general nonzero. The expansion reads $\partial_\gamma F(1, \tau) = V_0'(1) + \tau^2 G'(1) + O(\tau^4)$, and for small τ the population slope decides: if $V_0'(1) < 0$ then full coupling is a strict local minimizer whatever the sign of $G'(1)$, and if $V_0'(1) > 0$ the minimizer of V_0 is away from full coupling already at $\tau = 0$. The displacement (17) is then a second-order correction to a zeroth-order misspecification effect, and the sign of $G'(1)$ matters only when $|V_0'(1)|$ is itself of order τ^2.*

The displacement in (17) is of second order in the noise, and the gain from making it is of fourth order. When $G'(1) > 0$,

$$F(1, \tau) - F(\tilde{\gamma}(\tau), \tau) = \frac{G'(1)^2}{2\, V_0''(1)}\, \tau^4 + O(\tau^6). \tag{18}$$

A large gain at finite noise should not be inferred from the local sign.

The structure of the bridge does not determine the sign of $G'(1)$. Both signs occur, and each is strict in the examples below, so each persists on a relatively open set within the gateway-model parameterization.

7.1 Identical clusters

Give each of k clusters a knot of unit variance and one independent asset of variance $1/\delta$, and let every pair of knots have correlation c. By corollary 2 this is less special than it looks. By symmetry a portfolio is described by one number, the total weight x on the knots, and with $L = 1 + (k - 1)c$ its variance is

$$V(x) = \frac{1}{k}\Big[L\, x^2 + \frac{(1 - x)^2}{\delta}\Big], \qquad V(x) - V(x_\star) = \frac{L + 1/\delta}{k}\,(x - x_\star)^2, \qquad x_\star = \frac{1}{1 + \delta L}. \tag{19}$$

The bridge is estimating the split between a correlated basket and an independent one.

If the estimated knot correlation is z, proposition 4 gives the member part δ and the knot exposure

$$t_\gamma(z) = \frac{1 + (k - 2)z - \gamma(k - 1)z}{1 + (k - 2)z - \gamma(k - 1)z^2}, \qquad \text{so} \qquad x_\gamma(z) = \frac{t_\gamma(z)}{t_\gamma(z) + \delta}. \tag{20}$$

No matrix inversion remains. For $0 < z < 1$ the exposure is strictly decreasing in γ, because $\partial_\gamma t_\gamma(z)$ has the sign of $-z(k-1)(1-z)(1+(k-2)z)$.

An exact interior optimum. Let the estimate be $Z \in \{0, 2c\}$ with equal probability, for some $0 < c < \frac{1}{2}$. On the branch $Z = 0$ damping does nothing. On the branch $Z = 2c$ the choice

$$\gamma_\star = \frac{1 + 2(k-2)c}{2\,[1 + (k-3)c]} \tag{21}$$

gives $t_{\gamma_\star}(2c) = 1/L$ and hence $x = x_\star$, the population optimum. One branch is constant and the other attains its absolute minimum, uniquely since t_γ is strictly monotone. So $\gamma_\star$ is the unique global optimum. It lies strictly inside $(0, 1)$ and does not depend on δ.

With two clusters, $c = \frac{1}{4}$ and $\delta = 1$ the optimum is $\gamma_\star = \frac{2}{3}$, and in exact arithmetic $F(0) - F(\frac{2}{3}) = 1/576$ and $F(1) - F(\frac{2}{3}) = 1/900$.

Both signs in one model. Now let the noise be small and symmetric, $Z = c \pm \tau$, with $k = 10$ and $c = \frac{1}{4}$. Then (17) becomes

$$\tilde{\gamma}(\tau) = 1 - \frac{8 - \delta}{1 + 13\delta/4}\,\tau^2 + O(\tau^4). \tag{22}$$

For $\delta < 8$ the optimum is interior for small noise, for $\delta > 8$ it stays at full coupling, and at $\delta = 8$ higher orders decide. Here V_0 has a unique minimizer, again because t_γ is monotone, and $F(\cdot, \tau) \to V_0$ uniformly, so by proposition 8 these conclusions are global on the bridge.

Take $\delta = 12$. Then $\tilde{\gamma} = 1 + \tau^2/10 + O(\tau^4)$ and the constrained optimum stays at 1 under small noise. At the larger noise level $\tau = \frac{1}{4}$, which is $Z \in \{0, \frac{1}{2}\}$, (21) gives $\gamma_\star = 10/11$. The same symmetric model sits at full coupling under small noise and moves inside under larger noise.

Why the sign can go either way. At full coupling $x_1(Z) = 1/(1 + \delta[1 + (k-1)Z])$, which is convex in Z. Unbiased correlation noise therefore biases the expected knot allocation upward. Damping toward NCO also raises the knot allocation when correlations are positive. It reduces sensitivity to the noise and can worsen that bias, and the sign of $G'(1)$ is the outcome of the competition.

7.2 Less symmetric examples

A quartic. Take clusters $\{1, 2\}$ and $\{3, 4\}$ with knots 1 and 3, unit variances except $\Sigma_{33} = 4$, a single nonzero covariance $\Sigma_{13} = \frac{1}{2}$ between the knots (correlation $\frac{1}{4}$), and the estimate $\widehat{\Sigma}_{13} = Z \in \{0, 1\}$ with equal probability. On the branch $Z = 1$ the second knot exposure is $x = \kappa_2(\gamma) = (1 - \gamma)/(4 - \gamma)$ and the population variance is $V(x) = (49x^4 - 7x^3 + 36x^2 - 2x + 6)/(2(7x^2 + 3)^2)$. Its derivative has the sign of $P(x) = 49x^4 + 84x^3 - 21x^2 + 48x - 6$, which is increasing on $[0, \frac{1}{4}]$ with $P(0) = -6$ and $P(\frac{1}{4}) = 1585/256$.

So there is a unique optimum $\gamma_\star \approx 0.5587$, strictly inside the bridge, with $F(0) - F(\frac{1}{2}) = 3/1210$ and $F(1) - F(\frac{1}{2}) = 5/1452$ in exact arithmetic. With small symmetric noise $Z = \frac{1}{2} \pm \tau$ the same covariance has $G'(1) = 31779128936/554523204167$ and $V_0''(1) = 48136/3571279$, and (17) predicts $\tilde{\gamma}(0.05) \approx 0.9894$ against a computed optimum of 0.9896.

Full coupling under asymmetric noise. Take three clusters, each a knot and one remaining member that is uncorrelated with everything, with member variances 1, 4 and 8. Let the knot covariance

and its estimates be

$$S = \begin{pmatrix} 5 & 6 & 5 \\ 6 & \frac{33}{2} & 12 \\ 5 & 12 & \frac{25}{2} \end{pmatrix}, \qquad \widehat{S}_\tau = S \pm \tau \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

the two signs equally likely. The noise is symmetric, touches only cross-knot entries, and preserves the gateway model. Exact differentiation gives $V_0''(1) \approx 2.3949$ and

$$G'(1) = -\frac{4720238484060245648386156800}{80690793926829447510292364025 1} \approx -0.0058498 < 0,$$

so by proposition 8 full coupling is a strict local minimizer for all sufficiently small τ. As a spot check, at $\tau = \frac{1}{10}$ and $\tau = \frac{1}{100}$ exact evaluation gives $F(\gamma, \tau) > F(1, \tau)$ for $\gamma \in \{0.999, 0.99, 0.9, 0.5, 0\}$, which is consistent with a global optimum there but does not prove one.

Neither end owns the bridge. An interior optimum is not a consequence of the architecture, and a statement that it is generic needs a noise model. The derivatives quoted in both examples are exact, computed by truncated bivariate jets in rational arithmetic.

Remark 10 (Whole-matrix noise)**.** *Propositions 7 and 8 allow any symmetric E. Only the examples restrict the noise to cross-knot entries that preserve the gateway model, which is what reduces each cluster to the scalar $t_\gamma(z)$ or $\kappa_i(\gamma)$ and makes $G'(1)$ a closed form. Under whole-matrix noise $\widehat{\Sigma}_\tau$ need not satisfy the gateway model even when Σ does, so proposition 4 no longer describes the estimate: the member parts $d_{i,J_i}(\gamma)$ computed from $\widehat{\Sigma}_\tau$ are random and in general move with γ. Proposition 8 still applies, since it needs only symmetric noise, the smoothness assumed above, and gateway exactness of the population, but $G'(1)$ then collects within-cluster and cross-channel terms that the examples do not contain, and its sign is not settled by them.*

8 Rank deficiency

Everything so far assumes $\Sigma \succ 0$. This section allows $\Sigma \succeq 0$.

A singular covariance is one of two kinds. If some null portfolio has a nonzero total, a fully invested portfolio of zero variance exists. If every null portfolio is self-financing, the redundant assets can be removed, and only then is the pseudoinverse formula $w_\star = \Sigma^\dagger \mathbf{1} / \mathbf{1}^\top \Sigma^\dagger \mathbf{1}$ valid. The condition is $\mathbf{1} \in \mathrm{range}(\Sigma)$. For $\Sigma = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$ the portfolio $(2, -1)$ has budget 1 and variance 0, while normalizing $\Sigma^\dagger \mathbf{1}$ gives $(\frac{1}{3}, \frac{2}{3})$ with variance $25/9$.

Harmless redundancy breaks the recipe. Take two clusters, each a knot and an independent asset, all of unit variance, and make the two knots duplicates. The budget vector lies in the range of Σ, the optimum is $w_\star = (\frac{1}{6}, \frac{1}{3}, \frac{1}{6}, \frac{1}{3})$ in the order knot, member, knot, member, and $V_\star = \frac{1}{3}$. For every $\gamma < 1$ the pairs are $Q_i = \mathrm{diag}(1 - \gamma, 1)$ and $b_i = (1 - \gamma, 1)$, so both directions are $(1, 1)$ and the bridge holds equal weights.

At full conditioning each knot is completely explained by the other, and each local pseudoinverse discards its own knot. Together they discard a common return that is still useful. The singular Schur equations leave some weights undetermined, and choosing a minimum-norm solution separately in each cluster does not coordinate those choices.

Proposition 5 explains the approach to this case. As a knot becomes redundant $r_i \to 0$, so at every fixed $\gamma < 1$ its effective damping tends to zero and all the movement is compressed into a

Construction	Variance
Global optimum	1/3
Bridge at every $\gamma < 1$	3/8
Bridge at $\gamma = 1$ with each Q_i^{-1} replaced by $Q_i^\dagger$	1/2

Table 2: Duplicated knots.

shrinking interval near $\gamma = 1$. The limits do not commute. With the knot covariance regularized to $\mathbf{1}\mathbf{1}^\top + \varepsilon I$,

$$\lim_{\varepsilon\downarrow 0}\lim_{\gamma\uparrow 1} V_0 = \tfrac{1}{3}, \qquad \lim_{\gamma\uparrow 1}\lim_{\varepsilon\downarrow 0} V_0 = \tfrac{3}{8}.$$

This is more than numerical instability. The parameterization by γ is itself singular there.

Proposition 9 (A bridge in effective exposure)**.** *Let the knot covariance $S \succeq 0$ have positive diagonal, let $\mathbf{1} \in \mathrm{range}(S)$, and let every $\delta_i > 0$. Put $h = S^\dagger \mathbf{1}$ and, for $\lambda \in [0, 1]$,*

$$\bar\kappa_i(\lambda) = (1-\lambda)\,\frac{1}{s_i} + \lambda\, h_i, \tag{23}$$

keep the member parts of proposition 4, and allocate across the resulting cluster portfolios as before. The outer matrix $\bar K S \bar K + \Delta$ is positive definite, the portfolio is continuous in λ, it is NCO at $\lambda = 0$, and it is minimum variance at $\lambda = 1$.

Proof. Positive definiteness follows from $\Delta \succ 0$. At $\lambda = 1$, $Sh = \mathbf{1}$ gives $\big(\mathrm{diag}(h)\, S\, \mathrm{diag}(h) + \Delta\big)\mathbf{1} = h + \delta$, so the outer solution is $a = \mathbf{1}$ and the portfolio has knot exposures proportional to h and member weights proportional to $E_i^{-1}(\mathbf{1} - \beta_i)$, which is the minimum-variance portfolio of the compressed problem. □

For duplicated knots $\bar\kappa_i = 1 - \lambda/2$, and the portfolio moves continuously from variance 3/8 to 1/3. When $S \succ 0$ this path gives every cluster the same effective damping λ, which corresponds to cluster-specific parameters $\gamma_i = \lambda/(r_i + (1 - r_i)\lambda)$. It is a different interior path with the same two ends, and we do not claim it performs better out of sample. It makes a design choice visible: whether clusters should share a nominal γ or an effective λ.

Remark 11 (Noise around a singular covariance)**.** *If Σ is singular and both $\Sigma + \tau E$ and $\Sigma - \tau E$ are positive semidefinite for some $\tau > 0$, then $E \ker(\Sigma) = 0$. For $v \in \ker \Sigma$ both signs force $v^\top E v = 0$, and a positive semidefinite matrix with $v^\top(\Sigma + \tau E)v = 0$ has $(\Sigma + \tau E)v = 0$. Symmetric additive noise must therefore preserve the structural null space. A singular sample estimate of a positive definite population covariance is a different problem, since its empirical null directions are not riskless. In proposition 8 the condition $\dot w \neq 0$ becomes $\dot w^\top \Sigma \dot w > 0$.*

9 Cost

No step forms an $n \times n$ inverse. The largest linear solve has dimension $\max(k, \max_i |I_i|)$, as in NCO: the outer tier is $k \times k$ and each conditioning inverse is $(k-1) \times (k-1)$. The bridge adds work that NCO does not do: the cross-block products $\Sigma_{i,P_{-i}}$, and a leave-one-knot-out inverse per cluster, obtainable from Σ_{PP}^{-1} by a rank-one downdate.

If the gateway model fails, (5) approximates the full complement in (3), and Δ_i in (7) is what is lost.

10 Closing

Nested clustered optimization has been treated as belonging to a different family from optimization proper: a clustering heuristic that quarantines the instability of Markowitz rather than a relaxation of it. This paper shows otherwise. Under a rank-one model of cross-cluster dependence, a Schur relaxation places NCO and the global minimum-variance portfolio at the two ends of a one-parameter family of portfolios, at a cost no greater than NCO's own.

The family has structure. Along it only the knots move, each by an explicit fraction of its journey, and the variance given up at every setting is in closed form. It also comes with conditions for where to sit under estimation error. The sign of one derivative at the minimum-variance end decides whether remaining at that endpoint is optimal out of sample, and a symmetric family exhibits both outcomes in a single model, with the optimal coupling exact.

The separation reasserts itself when the knot covariance is rank deficient. There the parameterization by γ is itself singular, and NCO and the optimum stay apart on it, though a path in effective exposure still joins the two ends. The ingredients throughout are old: block inversion, the truncation of a conditioning set as in Vecchia approximations, and the damping of [2].

Two directions are open. Proposition 1 may serve Schur-complementary allocation on the tree, where the complement against a sibling block could be taken against that block's knots. The whole-matrix analysis of [7] rests on a scalar reduction specific to the tree and needs a replacement here. We leave both, and an empirical comparison, to future work.

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